NATURAL FREQUENCIES OF A THIN CONICAL SHELL

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Summary The natural frequencies of a thin conical shell are calculated under the assumption that no vibrations cause expansion of the shell. The calculation method is from Lord Rayleigh [5, §235 f], [4, page 551], [3]. To approximate the conditions practically encountered in loudspeaker cones, it is assumed that the conical shell is enclosed on one side by a rigid disk and on the other side free. The transition from the cone to the known formulas for the cylinder is carried out. A simple expression for the natural frequencies is obtained asymptotically.

1. Introduction

The moving part (diaphragm) of loudspeakers is often designed in the shape of a cone for reasons of stiffness.

The differential equations for the deformation of a conical shell [1, page 603] are so complex that no attempt has been made to calculate the natural frequencies based on these equations. In contrast, the calculation using Rayleigh's method [4] for a thin shell (which applies to loudspeakers) is straightforward.

If coordinates are introduced on the surface (x in the generating direction towards the apex), y in the tangent direction perpendicular to the axis, $\bar{z}$ in the direction perpendicular to the generating direction and inwards to y, then before deformation:

$$\bar{z} = \frac{1}{2}\left(\frac{x^2}{\rho_1} + \frac{y^2}{\rho_2}\right).$$

with ρ_1 and ρ_2 the principal radii of curvature ($\rho 1 = \infty$).

After deformation, however:

$$\bar{z} = \frac{1}{2}\left(\frac{x^2}{\rho_1 + d\rho_1} + \frac{y^2}{\rho_2 + d\rho_2} + 2\tau xy\right).$$

Suppose that $2d$ is the shell thickness, E is the modulus of elasticity, σ is the Poisson constant.

$$E = \frac{-3n}{m - n/3}, \quad \sigma = \frac{(m-n)}{2m}.$$

If $\delta(1/\rho_1)$, $\delta(1/\rho_2)$, and τ are known, the potential energy per unit area of the shell is written as:

$$U = \frac{2nd^2}{3}\left(\left(\delta\frac{1}{\rho_1}\right)^2 + \left(\delta\frac{1}{\rho_2}\right)^2 + \frac{m-n}{m+n}\left(\delta\frac{1}{\rho_1} + \delta\frac{1}{\rho_2}\right)^2 + 2\tau^2\right). \tag{1}$$

Integration over the shell area yields the total potential energy. If the kinetic energy is also known, the natural frequencies can be calculated.

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2. Calculation of Potential Energy

The first step is to express the aforementioned surface coordinates in cylindrical coordinates, measured from the cone apex, and r and φ measured from the considered lateral surface point $(r_o, z_o, 0)$.

The formulas are (β = angle axis-generating line):

$$\begin{aligned} x &= \frac{r_o}{\sin(\beta)} - r\cos(\beta)\sin(\beta) - z\cos(\beta) \\ y &= r\sin(\varphi) \\ \bar{z} &= 2\sin(\beta) - r\cos(\varphi)\cos(\beta) \end{aligned} \tag{2}$$

Near to the point $(r_o, z_o, 0)$ (except for second-order terms in φ and $z - z_o$)

$$\begin{aligned} x &= \frac{r - r_o}{\sin(\beta)} = \frac{z - z_o}{\sin(\beta)} \\ y &= r\varphi \\ \bar{z} &= 0. \end{aligned} \tag{2a}$$

If r, z, φ each receive a small increment u, v, w, the following equations result:

$$\begin{aligned} x = {} & \frac{r_o}{\sin(\beta)} - z_o\cos(\beta) - r_o\cos(\beta)\sin(\beta) - [u_o\sin(\beta)] \\ & - \left(\frac{du}{dz}\right)_o (z - z_o)\sin(\beta) - \left(\frac{du}{d\varphi}\right)_o \varphi\ \sin(\beta) - [v_o\cos(\beta)] \\ & - \left(\frac{dv}{dz}\right)_o (z - z_o)\cos(\beta) - \left(\frac{dv}{d\varphi}\right)_o \varphi\cos(\beta) + [r_o w_o \varphi\sin(\beta)]; \\ u_o w_o = {} & (r - r_o)w_o = (z - z_o)\tan(\beta)w_o; \end{aligned} \tag{3}$$

Equation 3 continues on the next page.

$$\begin{aligned} y &= r_0\varphi + [r_0 w_0] + u_0\varphi + r_0\left(\frac{dw}{dz}\right)_0 (z - z_0) + r_0\left(\frac{dw}{d\varphi}\right)_0 \varphi; \\ \bar{z} &= z_0\sin\beta - r_0\cos\beta + [v_0\sin\beta] - [u_0\cos\beta] \\ &\quad + \left(\frac{dv}{dz}\right)_0 (z - z_0)\sin\beta + \left(\frac{dv}{d\varphi}\right)_0 \varphi\sin\beta \\ &\quad - \left(\frac{du}{dz}\right)_0 (z - z_0)\cos\beta - \left(\frac{du}{d\varphi}\right)_0 \varphi\cos\beta + r_0 w_0\varphi\cos\beta. \end{aligned} \tag{3}$$

Terms of the second and higher order in φ and $z - z_o$ are neglected here. The subscript 0 indicates that the relevant quantities are formed at the point, $r = r_o$, $z = z_o$, $\varphi = 0$.

The surface can now be returned to its original position (neglecting terms of second and higher order) by means of an infinitesimal translation and rotation. The translation is equal and opposite to the terms enclosed in square brackets in (3).

The additional rotational terms for x, y, and $\bar{z}$ are, respectively:

$$r_o\phi\omega_3; \quad (z - z_o)\frac{\omega_3}{\cos\beta}; \quad -(z - z_o)\frac{\omega_3}{\cos\beta} - r_o\phi\omega_1$$

A necessary condition for regaining the zero position described above is that,

$$r\omega_3 - \frac{du}{d\varphi}\sin\beta - \frac{dv}{d\varphi}\cos\beta + rw\sin\beta = 0; \tag{4a}$$

$$\frac{du}{dz}\tan\beta + \frac{dv}{dz} = 0; \tag{4b}$$

$$u \quad + r\frac{dw}{d\varphi} = 0; \tag{4c}$$

$$w\tan\beta + r\frac{dw}{dz} + \frac{\omega_3}{\cos\beta} = 0; \tag{4d}$$

$$rw\cos\beta + \frac{dv}{d\varphi}\sin\beta - \frac{du}{d\varphi}\cos\beta - r\omega_1 = 0; \tag{4e}$$

$$-\frac{dv}{dz}\sin\beta + \frac{du}{dz}\cos\beta + \frac{\omega_2}{\cos\beta} = 0. \tag{4f}$$

(Index 0 omitted) Eliminating ω_3 from (4d) and (4a) yields:

$$r^2\frac{dw}{dz} + \frac{du}{dq}\tan\beta + \frac{dv}{d\varphi} = 0 \tag{4ad}$$

It is easy to verify that equations (4b), (4c), and (4ad) are identical to those obtained by Lord Rayleigh [5, p 399 Eq 14] by a different method. Their solution, under the condition that the cone's surface is enclosed by an inextensible disk at $z = l_1$, $u(x, y, l_1) = 0$ and $w(x, y, l_1) = 0$, is as follows: For $i \geq 2$,

$$\begin{aligned} u &= \sum_i i\tan\beta\,(z - l_1)\,A_i\sin i\varphi; \\ v &= \sum_i -\tan^2\beta\left(\frac{l_1}{i} + i\,(z - l_1)\right)A_i\sin i\varphi, \\ w &= \sum_i\left(1 - \frac{l_1}{z}\right)A_i\cos i\varphi. \end{aligned} \tag{5}$$

To calculate the potential energy according to Equation (1), it is necessary to take into account all second-order terms in ϕ and $z - z_0$ in the final equation (3) (after performing the infinitesimal rotation described above).

Taking ω_1 and ω_2 from Equations (4e) and (4f) respectively,

$$\begin{aligned} 2\bar{z} = &\left\{\frac{d^2v}{dz^2}(z - z_0)^2 + \frac{d^2v}{d\varphi^2}\varphi^2 + 2\frac{d^2v}{d\varphi}(z - z_0)\,\varphi\right\}\sin\beta \\ &- \left\{-u\varphi^2 - r\varphi^2 + \frac{d^2u}{dz^2}(z - z_0)^2 + 2\frac{d^2u}{dzd\varphi}(z - z_0)\,\varphi\right. \\ &\left.+\frac{d^2u}{d\varphi^2}\varphi^2 - 2r\frac{dw}{dz}(z - z_0)\,\varphi - 2r\frac{dw}{d\varphi}\varphi^2\right\}\cos\beta \\ &+ 2\frac{dv}{dz}\varphi^2\sin^2\beta\cos\beta - 2\frac{du}{dz}\varphi^2\sin\beta\cos^2\beta. \end{aligned}$$

Using $x = (z - z_o)/\cos\beta$, $y = r\phi$, it follows from this that:

$$
\begin{aligned}
2\bar{z} = &- 2xy\frac{\cos\beta}{r}\left\{\frac{d^2v}{dzd\varphi}\sin\beta - \frac{d^2u}{dzd\varphi}\cos\beta + r\frac{dw}{dz}\cos\beta\right\} \\
&+ \frac{y^2}{r^2}\left\{\frac{d^2v}{d\varphi^2}\sin\beta + u\cos\beta + r\cos\beta - \frac{d^2u}{d\varphi^2}\cos\beta\right. \\
&\left.+2r\frac{dw}{d\varphi}\cos\beta - r\frac{du}{dz}\sin\beta\cos^2\beta + r\frac{dv}{dz}\sin^2\beta\cos\beta\right\} \\
&+ x^2\cos^2\beta\left\{\frac{d^2v}{dz^2}\sin\beta - \frac{d^2u}{dz^2}\cos\beta\right\}.
\end{aligned}
$$

Taking into account Equations (4b), (4c), and (4ad), one can simplify to:

$$
(6) \qquad \begin{cases}
2\bar{z} = - 2xy\dfrac{\cos\beta}{r}\left\{\dfrac{d^2v}{dzd\varphi}\sin\beta - \dfrac{d^2u}{dzd\varphi}\cos\beta + r\dfrac{dw}{dz}\cos\beta\right\} \\
\quad + \dfrac{y^2}{r^2}\left\{\dfrac{d^2v}{d\varphi^2}\sin\beta + r\cos\beta - u\cos\beta - \dfrac{d^2u}{d\varphi^2}\cos\beta\right. \\
\quad \left. -r\dfrac{du}{dz}\sin\beta\cos^2\beta + r\dfrac{dv}{dz}\sin^2\beta\cos\beta\right\}.
\end{cases}
$$

The transition to the cylinder $beta = 0$ leads to Lord Rayleigh's formulas [5, p 414 Eq 6].

Before deformation, (6) read:

$$2\bar{z} = y^2\frac{\cos\beta}{r^2}.$$

The quantities $\delta\frac{1}{e_1}$, $\delta\frac{1}{e_2}$ and r in Equation (1) are thus:

$$
(7) \qquad
\begin{aligned}
\delta\left(\frac{1}{\rho_1}\right) =& 0; \\
\delta\left(\frac{1}{\rho_2}\right) =& \frac{1}{r^2}\frac{d^2v}{d\varphi^2}\sin\beta - \frac{d^2u}{d\varphi^2}\frac{\cos\beta}{r^2} - \frac{u\cos\beta}{r^3} \\
& - \frac{\sin\beta\cos^2\beta}{r}\frac{du}{dz} + \frac{\sin^2\beta\cos\beta}{r}\frac{dv}{dz}; \\
-\tau =& \frac{\cos\beta}{r}\left\{\frac{d^2v}{dzd\varphi}\sin\beta - \frac{d^2u}{dzd\varphi}\cos\beta + r\frac{dw}{dz}\cos\beta\right\}.
\end{aligned}
$$

Substituting the solutions (5) into (7) and the resulting expressions into (1) after integration over $0 \le \varphi \le 2\pi$ yields:

(8)
$$\begin{aligned}
\delta\left(\frac{1}{\varrho_2}\right) &= \sum_i A_i \sin(i\varphi)\ \alpha_i \left(\frac{1}{z} - \frac{l_1}{z^2}\right);\\
\alpha_i &= \frac{i^3 - i}{\sin\beta};\\
-\tau &= \sum_i A_i \cos(i\varphi)\ \frac{1}{z}\cdot\left(\frac{l_1}{z}\cos^2\beta - i^2\right);\\
U &= \frac{4\pi n d^2}{3}\sum_i A_i^2 \left\{\frac{m}{m+n}\alpha_i^2\left(\frac{1}{z} - \frac{l_1}{z^2}\right)^2 + \frac{1}{z^2}\left(\frac{l_1}{z}\cos^2\beta - i^2\right)\right\}.
\end{aligned}$$

From this, the total potential energy is obtained by integration over z:

(9)
$$\begin{aligned}
V &= \int_{l_1}^{l_2} dz\cdot z\tan\beta\cdot U\\
&= \frac{4\pi n d^3}{3} i\tan\beta \sum_i A_i^2 \left[\frac{m\alpha_i^2}{m+n}\left\{\ln\frac{l_2}{l_1}\right.\right.\\
&\left. +2\left(\frac{l_1}{l_2} - 1\right) - \frac{1}{2}\left(\frac{l_1^2}{l_2^2} - 1\right)\right\} - \left(\frac{l_1^2}{l_2^2} - 1\right)\frac{\cos^4\beta}{2}\\
&\left. +2i^2\cos^2\beta\left(\frac{l_1}{l_2} - 1\right) + i^4\ln\frac{l_2}{l_2}\right].
\end{aligned}$$

This expression applies to static problems.

3. Natural Frequencies

The cone's surface has kinetic energy,

$$T = \varrho d \int_{l_1}^{l_2} dz\ z\tan\beta \int_0^{2\pi} d\varphi \left\{\left(\frac{du}{dt}\right)^2 + \left(\frac{dv}{dt}\right)^2 + \left(\frac{rdw}{dt}\right)^2\right\}$$

where ρ is the density. Assuming temporal variaiont of u, v, w of $\cos\omega t$:

(10)
$$\begin{aligned}
-T &= \pi\omega^2\varrho\ d\ l_1^4\tan^3\beta\cdot\pi\sum_i A_i^2\left[(i^2+1)\left\{\frac{1}{4}\left(\frac{l_2^4}{l_1^4} - 1\right)\right.\right.\\
&\left. +\frac{1}{2}\left(\frac{l_2^2}{l_1^2} - 1\right) - \frac{2}{3}\left(\frac{l_2^3}{l_1^3} - 1\right)\right\} + \tan^2\beta\left\{\frac{1}{2i^2}\left(\frac{l_2^2}{l_1^2} - 1\right)\right.\\
&+\frac{i^2}{4}\left(\frac{l_2^4}{l_1^4} - 1\right) + \frac{i^2}{2}\left(\frac{l_2^2}{l_1^2} - 1\right) + \frac{2}{3}\left(\frac{l_2^3}{l_1^3} - 1\right)\\
&\left.\left. -\left(\frac{l_2^2}{l_1^2} - 1\right) - \frac{2i^2}{3}\left(\frac{l_2^3}{l_1^3} - 1\right)\right\}\right].
\end{aligned}$$

From equations (9) and (10), the desired angular frequencies ($\omega = 2\pi$ Hz) are immediately obtained:

$$\omega_i^2 = \frac{4nd^2}{3\varrho l_i^4 \tan^2\beta} \cdot \frac{F_i}{G_i} : \tag{11}$$

$$\begin{aligned} F_i = {} & \frac{m\alpha_i^2}{m+n}\left\{\ln\frac{1}{x} + 2(x-1) - \frac{1}{2}\left(x^2-1\right)\right\} \\ & - \frac{\cos^4\beta}{2}\left(x^2-1\right) + 2i^2\cos^2\beta\left(x^2-1\right) + i^4\ln\frac{1}{x}; \end{aligned} \tag{12}$$

$$\begin{aligned} G_i = {} & \left(i^2+1\right)\left\{\frac{1}{4}\left(\frac{1}{x^4}-1\right) + \frac{1}{2}\left(\frac{1}{x^2}-1\right)\right. \\ & - \frac{2}{3}\left(\frac{1}{x^3}-1\right) + \tan^2\beta\left\{\frac{1}{2i^2}\left(\frac{1}{x^2}-1\right) + \frac{i^2}{4}\left(\frac{1}{x^4}-1\right)\right. \\ & + \frac{i^2}{2}\left(\frac{1}{x^2}-1\right) + \frac{2}{3}\left(\frac{1}{x^3}-1\right) - \left(\frac{1}{x^2}-1\right) \\ & \left. - \frac{2i^2}{3}\left(\frac{1}{x^3}-1\right)\right\} \end{aligned} \tag{13}$$

From (8), for $i \geq 2$, $\alpha_i = \frac{i^3-i}{\sin\beta}$, and $x = l_1/l_2$.

In the case of a cylinder of radius R, $l_1 \tan\beta \sim l_1 \sin\beta = R$. This results in:

$$\omega_i^2 = \frac{4mnd^2}{3\rho R^4(m+n)}\left(\frac{\left(i^3-i\right)^2}{i^2+1}\right)\lim_{x\to 1}\frac{\ln\frac{1}{x} + 2(x-1) - \frac{1}{2}\left(x^2-1\right)}{\frac{1}{4}\left(\frac{1}{x^4}-1\right) + \frac{1}{2}\left(\frac{1}{x^2}-1\right) - \frac{2}{3}\left(\frac{1}{x^3}-1\right)}, \tag{14}$$

or,

$$\omega_i^2 = \frac{4d^2}{3\varrho R^4}\frac{mn}{m+n}\frac{\left(i^3-i\right)^2}{i^2+1},$$

a well-known formula [5, p 417 Eq 18].

It is often useful to have a simple formula at hand for high frequencies. From (11), (12), (13), and (8), we obtain:

$$\lim_{i\to\infty}\omega_i^2 = \frac{4d^2}{3\varrho l_i^4\tan^4\beta}\frac{mn}{m+n}i^4\frac{\ln\frac{1}{x} + 2(x-1) - \frac{1}{2}\left(x^2-1\right)}{\frac{1}{4}\left(\frac{1}{x^4}-1\right) + \frac{1}{2}\left(\frac{1}{x^2}-1\right) - \frac{2}{3}\left(\frac{1}{x^3}-1\right)} + \mathcal{O}(i^2)$$

I would like to point out that the calculation method used does not allow us to handle the case of a cone shell extending to the apex.

If the cone is held at l_2 instead of l_1 by a non-extensible disk, Equation (5) must be simply modified, which also changes (12) and (13) somewhat. It is left to the reader to handle this and possibly other special cases using the formulas above.

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